

SAI POTHAN JANJANAM

ML Engineer | GenAI | Computer Vision | ML Systems

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Location: India (Willing to relocate)

SUMMARY

ML Engineer with 5+ YOE developing ML systems across anomaly detection, computer vision, video analytics, and GenAI applications. Skilled in model training, inference, optimization, API design, and preparing production-ready services using PyTorch, Keras, FastAPI, Docker, and Azure. Curious and driven to solve challenging problems.

SKILLS

- **Programming:** Python, C, C++, Cython
- **Core ML & DL:** Supervised & Unsupervised Learning, CNNs, Encoder-Decoder Models, Anomaly Detection, Image Processing, Computer Vision, Video Analytics, Time Series, TTS
- **GenAI & LLM Tooling:** LLMs (OpenAI GPT-4/4o, Gemini, Claude, Open-source), Transformers, Embeddings, RAG, Multimodal RAG, LangChain, LlamaIndex, Rerankers (Cohere, Jina AI, Voyage AI), Prompt Engineering
- **Frameworks:** PyTorch, Keras, Scikit-Learn, FastAPI, OpenCV, Ollama, HuggingFace
- **Databases & Vector Search:** SQL, Milvus, LanceDB, Pinecone, KDB.ai (Contributor)
- **Systems, MLOps & Cloud:** Linux, Docker, AWS, Azure, MLflow, Git, Model Deployment, ROS

EXPERIENCE

- **Konverge.ai** Nagpur, IN
Assistant Lead Software Engineer (DS & AI) Sep 2020 – Present
 - KDB.ai Vector Database – Feature Development**
 - Developed code examples demonstrating the use of KDB.ai for LLM tasks, including Multimodal RAG, Hybrid Search, and a Sub-Question Sub Query Engine to support developer workflows.
 - Conducted an in-depth evaluation of existing search methods and showed that integrating a Reranker improved performance by **35%** on MS MARCO and Wiki QnA datasets.
 - Designed, developed, documented, tested, and shipped the Reranker feature with integrations for Cohere, Jina AI, and Voyage AI APIs - now part of KDB.ai feature set.
 - Fee Erosion Estimation System (Construction Analytics)**
 - Standardized and refactored the application by introducing scheduled caching, reducing initial load time by $\approx$ **33%** and reducing page-switching time by $\approx$ **83%**.
 - Developed and integrated the project schedule metrics and their correlation with project costing providing actionable insights about construction progress.
 - Designed a machine learning system combining time-series forecasting and multivariate time-series classification to predict the likelihood of project fee erosion.
 - ML-based Anomaly Detection Platform for Industrial Machines**
 - Developed an end-to-end anomaly detection platform for industrial machines - covering authentication, data processing pipelines, model training/retraining, inference, alerting, failure handling and job queuing.
 - Optimized an ML platform to replace an enterprise ML application, supporting model training on up to **6 months of data** with **150** features at **1-minute** resolution, queuing up to 5 models, and enabling near-real-time inference across **50** deployed models.
 - Talking Head Generation from Image & Text**
 - Led the end-to-end development of a talking-head video generation pipeline from a single image and text.
 - Integrated Text-to-Speech, lip synchronization, face reenactment and image enhancement modules and optimized the system to achieve an average of **12 FPS** of video generation.
 - Smart City - Video Analytics**
 - Optimized computer vision modules to achieve FPS improvements - Zone Violation by **40%**, Line Crossing by **25%**, Object tracking by **75%**.
 - Developed and deployed modules for Collision Detection, Collision Prediction, Object Tracking, Fire Detection, Social Distance Violation in real-time video analytics engine.
- **Swaayatt Robots** Bhopal, IN
Research Associate Sep 2019 – Feb 2020
 - Implemented a object detection model (YOLO-v3) to identify obstacles and non-obstacles on the road.
 - Worked on simulation of an autonomous vehicle for data collection using Gazebo & ROS.
 - Worked on mobile robot navigation to retrieve robot state information from MAVSDK.

RECOGNITION & IMPACT

- **Above and Beyond Award (Employee of the Year, Konverge.ai (2023)):** Recognized for enhancing the interview process, delivering consistent results in projects, collaborating effectively across teams, and contributing to client-facing solution design.
- **Impact on organisation's growth:** Interviewed **300+** candidates and contributed towards building an internal coding-question repository used in hiring process.

PROJECTS

Kriya - QA Automation Chatbot

- Built an organisation-wide chatbot that generates test scenarios, steps, and code across domains (Data Engineering, ML, UI/UX, APIs), improving efficiency of testing teams.
- Designed the underlying ML pipeline using Ollama (Gemma 3) with a FastAPI backend and deployed the system on Azure VM.

Opensource contributions to KDB.ai vector database

- Identified and resolved schema-related warnings in KDB.ai's hybrid search functionality caused by duplicated sparseVectors fields returned from backend. (GitHub PR)
- Updated the ingestion logic to normalize backend schemas, eliminating the warnings and improving data insertion reliability.

Confluence Docs Chatbot Application

- Developed a chatbot application that fetches documents from Confluence and answers queries using the LLM-RAG mechanism.
- Implemented key components: Extractor (Bulk, Incremental), Data Processor (Preprocessing, Chunking, Vector Embeddings, Vector Databases), LLM-RAG question-answering layer.
- Built using Streamlit frontend with fully extensible architecture supporting custom data processors, embeddings, and LLM models(online, self-hosted).

EDUCATION

- **VNIT Nagpur** Nagpur, India
B.Tech in Mechanical Engineering *June 2015 – May 2019*

CERTIFICATIONS & MOOCS

- **Machine Learning:** Coursera (Andrew NG)
- **Deep Learning Specialization:** Coursera (Andrew NG)
- **Applied Data Science:** WorldQuant University

LANGUAGES

- **English:** Professional Proficiency (C1)
- **Japanese:** Limited Working Proficiency
- **Hindi:** Bilingual Proficiency
- **Telugu:** Native Proficiency